



PTT PHENOL COMPANY LIMITED

## PROCESS SPECIFICATION

PREFLASH COLUMN FEED  
FILTERS (X-2302A/B)

Docu. No. : 14780-8120-PS-X2302

Project No. : 120117

Rev. No. : Z1

Date : MAY 10, 2016

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TITLE :

## PREFLASH COLUMN FEED FILTERS

(X-2302A/B)

DISCIPLINE : PROCESS

DOCUMENT NO. : 14780-8120-PS-X2302

PROJECT NAME : PTT Phenol Train II Project

PROJECT NO. : 120117

OWNER : PTT Phenol Company Limited (PPCL)

| Z1       | MAY 10, 2016      | AS BUILT            |              | S.W.Park  | J.S.Chung | S.W.Lee      |
|----------|-------------------|---------------------|--------------|-----------|-----------|--------------|
| F1       | MAY 29, 2014      | FINAL ISSUE         |              | S.W.Park  | J.S.Chung | S.W.Lee      |
| A1       | OCT 2, 2013       | APPROVED FOR DESIGN |              | S.W.Park  | J.S.Chung | S.W.Lee      |
| O1       | JUL 19, 2013      | COMMENT ISSUE       |              | S.W.Park  | J.S.Chung | S.W.Lee      |
| REV. NO. | DATE              | DESCRIPTION         |              | PREP'N    | REVIEW    | QA REVIEW    |
| REV. NO. | PREPARATION DEPT. | PREPARATION         | REVIEW       | QA REVIEW |           | APPROVAL     |
| Z1       | PROCESS           | S.W.Park            | J.S.Chung    | -         |           | S.W.Lee      |
|          |                   | MAY 10, 2016        | MAY 10, 2016 | -         |           | MAY 10, 2016 |

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**PROJECT SPECIFICATION**

963766 — 912

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**FILTERS**

| REV | DATE   | BY  | APVD | REV | DATE | BY | APVD |
|-----|--------|-----|------|-----|------|----|------|
| 0   | 9MAY12 | MYW | MEH  |     |      |    |      |
|     |        |     |      |     |      |    |      |
|     |        |     |      |     |      |    |      |
|     |        |     |      |     |      |    |      |

**UOP Phenol Process Unit  
Concentration, Decomposition and Direct Neutralization Sections**

|             |                                                |
|-------------|------------------------------------------------|
| Item Number | X-2302A/B<br>(P&ID No.: 14780-8120-25-23-0003) |
| Service     | Preflash Column Feed Filters                   |
| Quantity    | 2                                              |

Cartridge type filters for removal of solids from liquid streams. Filters shall meet the following requirements:

**Design and Operating Conditions**

| Operating Conditions                            |                                                                                                                   |
|-------------------------------------------------|-------------------------------------------------------------------------------------------------------------------|
| Fluid, mole%                                    | Oxidate: approximately 77.8% Cumene (isopropylbenzene) and 20.8% CHP (cumene hydroperoxide) with entrained salts. |
| Maximum Flow Rate, m <sup>3</sup> /hr           | 275                                                                                                               |
| Inlet Pressure, kg/cm <sup>2</sup> (g)          | 8.0                                                                                                               |
| Temperature, °C                                 | 82                                                                                                                |
| Fluid Specific Gravity @ Operating Temperature  | 0.845                                                                                                             |
| Fluid Viscosity, cP @ Operating Temperature     | 0.61                                                                                                              |
| Solid Materials to be Filtered out of Fluid     | Sodium salts of cumene hydroperoxide (CHP) and phenol; dirt, pipe scale, etc                                      |
| Consistency of Filter Cake                      | Jelly-like, containing particulate material                                                                       |
| Filter                                          |                                                                                                                   |
| Type                                            | Vendor to recommend                                                                                               |
| Nominal Particle Retention Rating, microns      | 10                                                                                                                |
| Maximum Clean Pressure Drop, kg/cm <sup>2</sup> | 0.35                                                                                                              |
| Maximum Dirty Pressure Drop, kg/cm <sup>2</sup> | 1.40                                                                                                              |

Other specifications  
referenced in this  
specification:

UOP Standard Specification: 3-11, 3-12

UOP Standard Drawing:

UOP Project Specification:

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**PROJECT SPECIFICATION**

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**FILTERS**

| REV | DATE   | BY  | APVD | REV | DATE | BY | APVD |
|-----|--------|-----|------|-----|------|----|------|
| 0   | 9MAY12 | MYW | MEH  |     |      |    |      |
|     |        |     |      |     |      |    |      |
|     |        |     |      |     |      |    |      |
|     |        |     |      |     |      |    |      |

| <b>Mechanical Design Conditions</b>         |                                       |
|---------------------------------------------|---------------------------------------|
| Pressure, kg/cm <sup>2</sup> g (Note 5)     | 11.5                                  |
| Temperature, °C                             | 120                                   |
| External Design pressure @ °C               | Full Vacuum @ 82                      |
| Minimum Design Metal Temperature (MDMT), °C | 15                                    |
| Inlet / Outlet Line Size, inches            | 8 / 8                                 |
| Housing Metallurgy/Internal                 | A-240 Type 304L Stainless Steel       |
| Filter Cartridge                            | Glass Fiber on Stainless Steel Core** |
| Corrosion Allowance, inches (mm)            | 1.5 mm                                |
| Flange Class and Rating                     | ASME CL150 RF                         |
| Drain Size, inches                          | 2                                     |
| Relief Valve Connection, inches             | 3                                     |

\* Deleted.

\*\* Materials must be compatible with cumene hydroperoxide at the design temperature.  
The following selections are acceptable (Or Equal):  
Pall – Ultiplex High Flow  
Rocky Mountain Filtration – Type C4VS or C4VT  
Parker Filtration – FULFLOW K15R40G304  
(Text or number underlined above varies with unit capacity)

**Notes:**

1. Filter housings shall be provided with connections for fire case relief valves.
2. Filter housings shall be provided with vents and low point drains.
3. Twenty-four spare elements are required for start-up.
4. Filter shells shall meet the requirements of ASME Boiler and Pressure Vessel Code, Section VIII, Division 1, Pressure Vessels. See also UOP Standard Specifications 3-11 and 3-12 for additional requirements.
5. Design Pressures are based on a pump shutoff pressure.

Revision  
Indication



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**FILTERS**

| REV | DATE   | BY  | APVD | REV | DATE | BY | APVD |
|-----|--------|-----|------|-----|------|----|------|
| 0   | 9MAY12 | MYW | MEH  |     |      |    |      |
|     |        |     |      |     |      |    |      |
|     |        |     |      |     |      |    |      |
|     |        |     |      |     |      |    |      |

**Manufacturer (Or Equal):**

Pall Corporation  
website: [www.pall.com](http://www.pall.com)

Amfilter B.V.  
website: [www.amfilter.nl/](http://www.amfilter.nl/)

Facet Enterprises, Inc  
Website: [www.facetinternational.com](http://www.facetinternational.com)

Pall Corporation  
Filterite Division  
website: <http://www.filterite.com/filterite/index2.htm>

Parker Filtration  
website: [www.parker.com](http://www.parker.com)

Rocky Mountain Filtration  
website: [www.rmfiltration.com](http://www.rmfiltration.com)

Faudi Filter Systems GmbH  
website: [www.faudi.de](http://www.faudi.de)

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